



Influence of Lighting and Gender

James Ma, Minh Nguyen, Ved Patel



Objective & Motivation

Objective

To analyze how gender and lighting conditions influence the performance of a feature-level fusion facial recognition system that uses geometric facial landmarks.

Why This Matters

- Facial recognition performance can vary across demographic groups.
- Illumination is one of the strongest factors affecting landmark quality.
- Feature-level models (non-deep-learning) are sensitive to geometric distortion.
- Understanding these effects helps identify early sources of bias and failure.

Research Questions (RQ)

RQ1:

Does gender (male vs female) significantly affect facial recognition accuracy when using geometric feature-level fusion?

RQ2:

How do different lighting conditions (low, medium, high brightness) influence recognition accuracy and discriminability in this system?



System Overview

Pipeline

- Facial landmark extraction
- Compute geometric features
 - Interocular distance
 - Lip area
- Compute lighting score (brightness)
- Similarity = normalized Euclidean distance
- Evaluate using EER, d' , ROC, DET

Dataset: Caltech Faces 1999 (450 images, 27 subjects)



Code Architecture

- **metadata_generator.py** → gender labels + brightness
- **analyze_face_features.py** → feature extraction
- **gender_test.py** → male vs female accuracy
- **lighting_test.py** → low/medium/high lighting accuracy
- **evaluator.py** → ROC + DET curves

Pipeline Steps

1. Generate metadata
2. Run gender test
3. Run lighting test
4. Produce plots (ROC/DET/Score distribution)



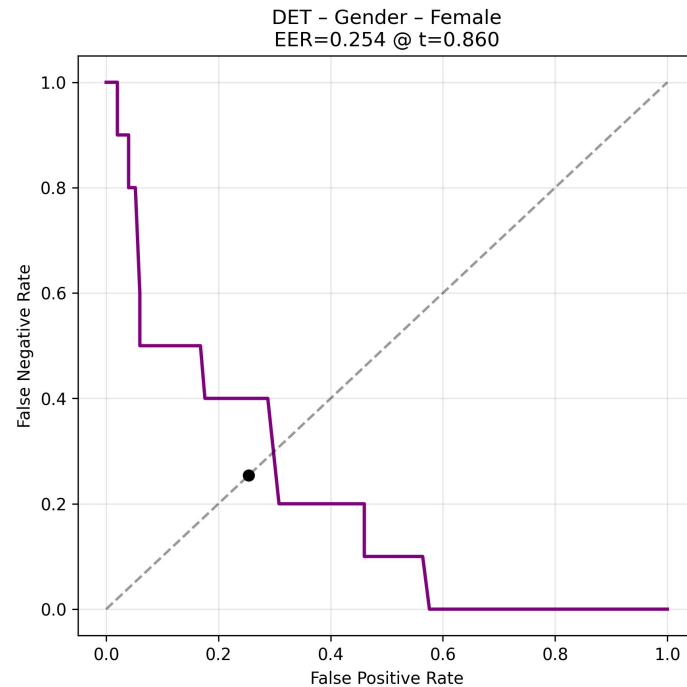
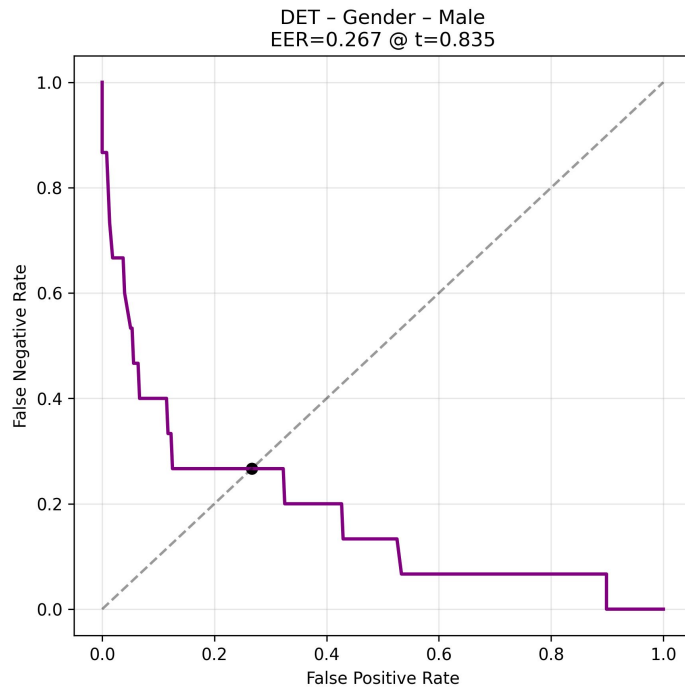
RQ1 — Gender Influence on Accuracy

Group	EER ↓	d' ↑	Interpretation
Male	0.267	1.149	Slightly higher error
Female	0.254	1.160	Marginally better separation

Key Points:

- Male and female EER values were almost the same.
- d' -prime scores were nearly identical.
- This shows gender had little to no impact on performance.

DET CURVES





RQ2 — Lighting Influence on Accuracy

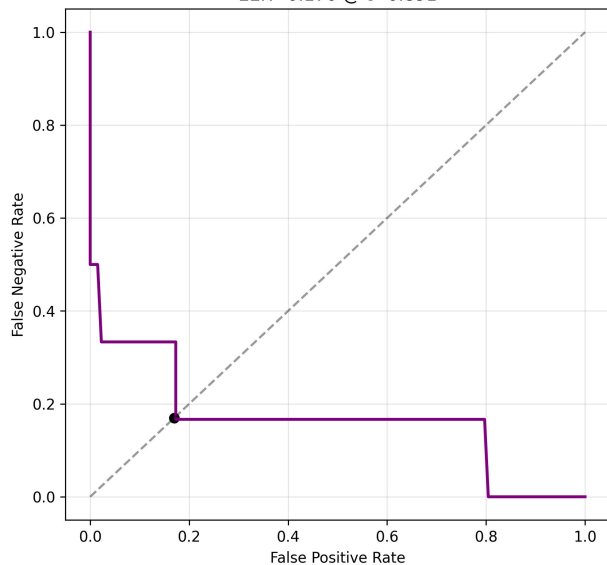
Lighting	EER ↓	d' ↑	Observation
Low	0.170	0.95	Poor edge detection
Medium	0.330	1.21	Best separation + accuracy
High	0.264	1.05	Glare distorts geometry

Key Points:

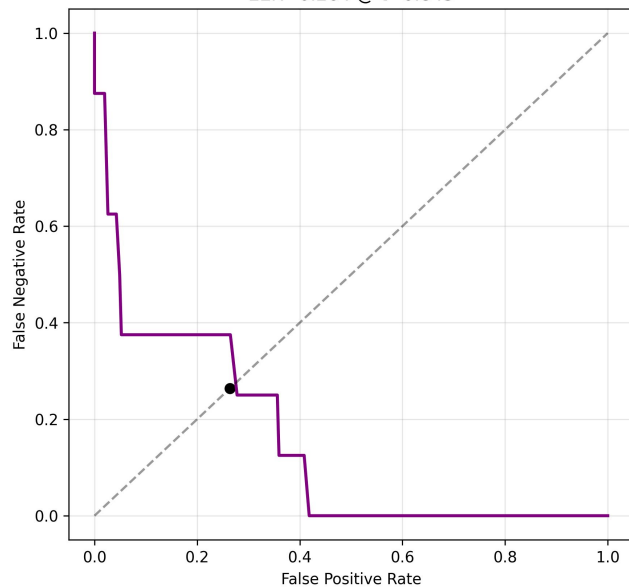
- Low and high lighting caused noticeable drops in accuracy.
- Medium lighting performed the best overall.
- Lighting had a much stronger effect on performance than gender.

DET CURVES

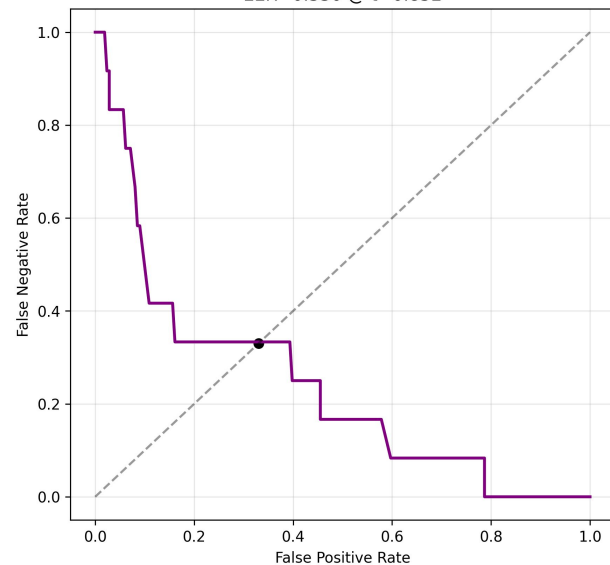
DET - Lighting - Low
EER=0.170 @ t=0.891



DET - Lighting - High
EER=0.264 @ t=0.843



DET - Lighting - Medium
EER=0.330 @ t=0.852



Why Lighting Matters



Low light → facial contours disappear

High light → specular highlights distort landmarks

Medium light = optimal balance

Larger effect on geometry than demographic traits



Ethics & Risks

Individual Risks

- Facial data is sensitive and uniquely identifying
- Potential for privacy misuse or unwanted surveillance

Societal Risks

- Bias amplification in real-world systems
- Misidentification in areas like law enforcement or hiring
- Erosion of public trust in biometric technology

Risk Mitigation (What We Did)

- Used only an academic, publicly available dataset
- No personal or user-generated images collected
- All processing done locally (no cloud storage)
- Clear, transparent reporting of bias-related results



Conclusion

Main Findings

- Gender: ~ 0.01 EER difference $\rightarrow$ **negligible bias**
- Lighting: large accuracy swings $\rightarrow$ **dominant factor**
- Medium lighting yields highest d'
- Feature-level methods can work but are **highly lighting-sensitive**

Future Work

- Add illumination normalization
- Use more diverse datasets
- Combine geometric + photometric features